

MICHAEL DANLEY

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EDUCATION

Duke University

B.S.E. in Electrical and Computer Engineering | B.S. in Computer Science
GPA: 4.0 / 4.0

Durham, NC
Expected May 2029

TECHNICAL SKILLS

Embedded: ESP-IDF, FreeRTOS, I²C, SPI, UART, RS-485, LoRa

Hardware: KiCad schematic capture and PCB layout, power electronics, Fusion 360, additive manufacturing, soldering

Software: Linux, Docker, Proxmox, ZFS, Git

Languages: Java, Python, C++

EXPERIENCE

Duke AERO

Lead Electrical & Controls Engineer, Liquid Propulsion Division

Durham, NC
August 2025 – Present

- Own the full electrical stack for a custom Engine Control Unit powering a 2.5 kN liquid rocket engine: architecture, schematic capture, PCB layout, and power distribution, plus high-frequency sensor sampling and closed-loop valve/actuator control in C++.
- Serve as lead electrical architect for Project Prometheus, an electric pump-fed bipropellant engine targeting a 2028 launch and the FAR-OUT challenge, directing a subteam across PCB design, embedded firmware, and real-time control software for pump control loops, active tank pressure regulation, and ground systems.

PORCH Communities

Instructional Design Intern

Carrboro, NC
May 2026 – August 2026

- Scoped and delivered an 8-module volunteer onboarding curriculum, supporting a nonprofit that mobilizes 1,500+ volunteers nationwide. Used Figma to design asynchronous interactive training experiences.
- Gathered requirements directly from active Neighborhood Coordinators and iterated the curriculum against their feedback, structuring content around measurable learning objectives and proposing a gamification system of badging, points, and leaderboards to improve retention.

Project ORCA, Cassar Lab

Lead Electrical Engineer

Durham, NC
August 2025 – May 2026

- Designed two custom PCBs in KiCad: a solar/battery-powered ESP32-C6 satellite module and a Raspberry Pi HAT chamber controller, integrating 13-channel spectral sensing (AS7343), GNSS positioning, a 24 V power tree (TPS5430 buck converters, BQ25185 near-MPPT charge management, XC6220 LDOs), and MOSFET-driven PWM for high-amperage LED modulation.
- Built a dual-path link: RS-485 over RJ45 carrying differential data and 12 V power for EMI immunity, with 915 MHz LoRa as a wireless option. Wrote the satellite firmware in C++ on ESP-IDF, accumulating spectral samples in RTC memory across deep-sleep cycles to hold draw at ~50 mA for unattended outdoor operation.
- Developed the chamber application in Python, including a real-time control loop mapping received spectral data to LED PWM, SQLite logging, a Flask dashboard, and an automated validation layer that models expected solar irradiance from GPS position and UTC time to flag physically implausible sensor readings.

Levita Health, Inc.

Design Engineering Intern

Durham, NC
February 2026 – April 2026

- Refined and validated mechanical assemblies for Uplift, a user-adjustable abdominal compression device, executing tensile stress testing and failure analysis to confirm structural integrity and using additive manufacturing to turn CAD concepts into testable models.

INDEPENDENT PROJECTS

Homelab Server

Jamestown, NC

- Deployed self-hosted services as Docker Compose stacks in dedicated Proxmox LXC containers with persistent data bind-mounted from ZFS and built the authentication layer around a self-hosted Dex OIDC provider bridging application login to Google OAuth, fronted by a Cloudflare Access email allowlist.
- Built the Proxmox VE host from salvaged hardware for roughly \$300, running a 3× 2 TB ZFS RAIDZ1 pool with per-dataset compression tuned to workload, hourly-through-monthly Sanoid snapshot retention, and weekly Vzdump backups, with ZED and smartd alerting over SMTP on pool degradation and SMART attribute trends before drives fail outright.